A MAP OF FLAT MODULES INJECTIVE MODULO THE MAXIMAL IDEAL PRESERVES CONTENT IDEALS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Editorial note. Benign-screen note: the proof contains the phrase 'it is clear that' / 'is obvious' once, used as a legitimate step inside a complete argument; the author proves the statement, so this is not a hint-only proof.

Allowed standard prerequisites (editorial).

- The content ideal of an element and its behaviour under inclusion of ideals. Definition \ref{definition-content-ideal}, cited at proof lines 5735-5737. Justifies the reduction to a comparison of ideals.
- The content ideal of an element is finitely generated, and Nakayama's lemma. Lemma \ref{lemma-content-finitely-generated} and Algebra, Lemma \ref{algebra-lemma-NAK}, cited at proof lines 5739-5741. Produce the nonzero map to the residue field.

Source reference. Original lemma 0ASC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13575–13575. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0ASA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13573–13573. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0ASB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13574–13574. Navigation pointer to a statement the

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Source reference. Original lemma 00DV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 512–512. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $(A, \mathfrak{m})$ be a local ring. Let $u : M \to N$ be a map of flat A-modules such that $\overline{u} : M/\mathfrak{m}M \to N/\mathfrak{m}N$ is injective. If $x \in M$ has content ideal I, then $u(x)$ has content ideal I as well.*

Proof. It is clear that $u(x) \in IN$. If $u(x) \in I'N$, then $u(x) \in (I' \cap I)N$, see discussion following Definition 0ASA. Hence it suffices to show: if $x \in I'N$ and $I' \subset I$, $I' \neq I$, then $u(x) \notin I'N$. Since I/I' is a nonzero finite A-module (Lemma 0ASB) there is a nonzero map $\chi : I/I' \to A/\mathfrak{m}$ of A-modules by Nakayama's lemma (Algebra, Lemma 00DV). Since I is the content ideal of x we see that $x \notin I''M$ where $I'' = \mathrm{Ker}(\chi)$. Hence x is not in the kernel of the map

$$IM = I \otimes_A M \xrightarrow{\chi \otimes 1} A/\mathfrak{m} \otimes M \cong M/\mathfrak{m}M$$

Applying our hypothesis on $\overline{u}$ we conclude that $u(x)$ does not map to zero under the map

$$IN = I \otimes_A N \xrightarrow{\chi \otimes 1} A/\mathfrak{m} \otimes N \cong N/\mathfrak{m}N$$

and we conclude. □